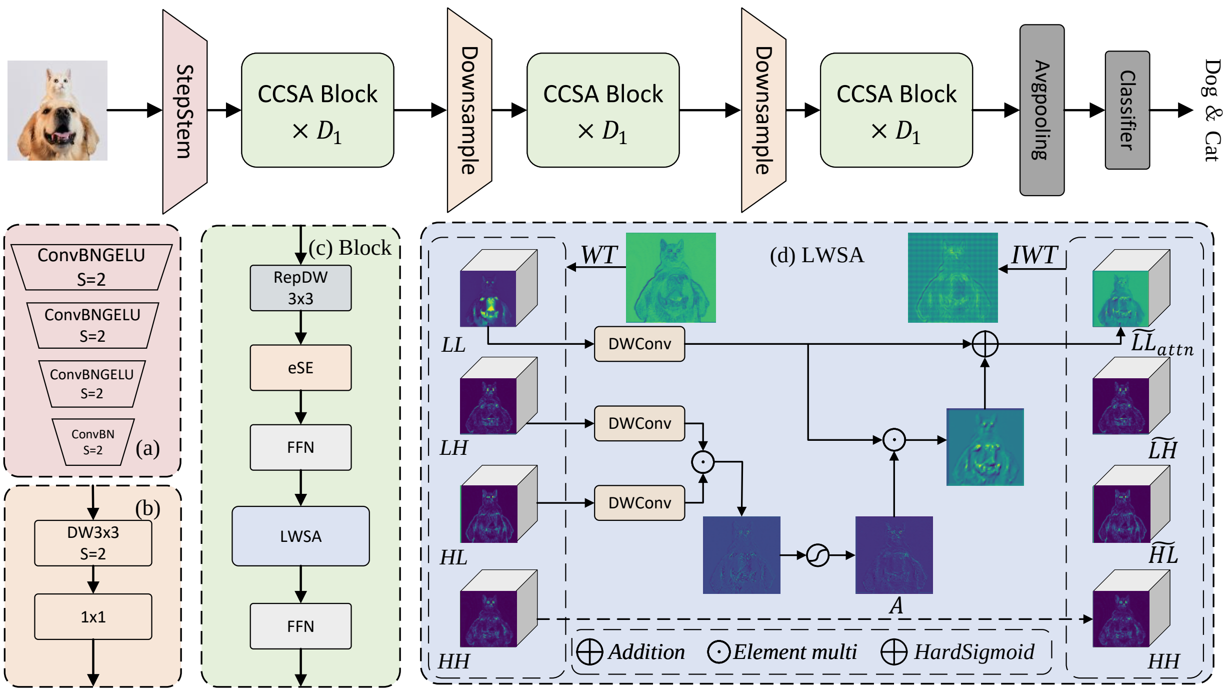
Graphical Abstract

WCANet: Effective Feature Modeling under Low Computational Cost

Fang Wang, Huitao Li, Wenhan Chao, Zheng Zhuo, Xinxin Yang



To design efficient neural networks, many studies focus on reducing redundancy. However, such feature discarding is often random and may remove useful information, which leads to degraded network performance. To address the challenge of insufficient feature extraction under limited computational budgets, we decouple feature modeling into channel-wise and spatial dimensions, which substantially reduces computational overhead. To enable efficient spatial modeling, we propose Learnable Wavelet Spatial Attention (LWSA), which leverages the wavelet transform to decompose the feature map into multiple subbands of varying informativeness and performs spatial attention selectively on the most informative components, enabling effective feature extraction at low computational cost. By cascading the ESE module and LWSA, we propose a Cascaded Channel–Spatial Attention mechanism for feature extraction. Building upon this, we further introduce WCANet, a lightweight architecture that achieves effective feature modeling under low computational budgets across a variety of vision tasks. For example, on ImageNet-1K, WCANet-T5 achieves approximately 42% higher throughput than FasterNet-T1 at comparable accuracy on GPU. WCANet-T4 also achieves 2.9× the throughput of StarNet-S1 with 1.4 percentage points higher Top-1 accuracy, and surpasses MobileMamba-T2 by 1.3 percentage points.

Highlights

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- To overcome the limitation that fixed wavelet filters cannot effectively decompose high-dimensional feature maps, we propose learnable wavelet filters with orthogonality constraints, allowing filters to adapt dynamically while strictly maintaining orthogonality.
- We propose a learnable wavelet spatial attention mechanism (LWSA). LWSA leverages the features from sub-bands with different orientations to perform spatial attention on informative sub-bands, thereby enhancing key spatial nodes and achieving efficient spatial feature modeling.
- We propose Cascade Channel–Spatial Attention (CCSA). CCSA utilizes ESE and LWSA to perceive key features at both channel and spatial levels, respectively, achieving an optimized accuracy-efficiency trade-off.
- We propose WCANet, a visual network design optimized for practical lightweight applications. Extensive experiments demonstrate that WCANet exhibits robust scalability and highly competitive performance across various devices.

WCANet: Effective Feature Modeling under Low Computational Cost

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ABSTRACT

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1. Introduction

With the proliferation of mobile devices, lightweight network designs have become a research hotspot in vision network design [1; 2; 3; 4; 5; 6]. Current mainstream lightweight models are primarily categorized into CNN-based, Transformer-based, and Mamba-based structures. Vision Transformers (ViTs) have drawn significant attention due to their global effective receptive field (ERF) and strong long-range dependency modeling capability. However, their quadratic computational complexity leads to substantially higher inference overhead compared to other architectures. Although some works [7; 8; 9; 10; 11; 12; 13] have been proposed to alleviate this issue with notable success, the computational bottleneck of ViTs still becomes evident under high-resolution inputs.

State-space models (SSMs) [14; 15; 16; 17] are able to capture long-range dependencies with linear computational complexity, motivating researchers to explore their application in vision tasks. Consequently, SSMs have been successfully adapted to the visual domain and have achieved notable results [18; 19]. MobileMamba [20], as a lightweight Mamba-based architecture, has achieved high throughput and high precision on the GPU. However, vision mambas (ViMs) are limited by GPU device support, and the inference latency on devices such as CPU will increase significantly. This severely restricts the performance of these methods in practical applications.

CNN-based methods [21; 11; 22] aim to achieve model lightweighting by reducing computational overhead~~redundancy~~. Early works [23; 24; 25; 26] leveraged DWConv to lower

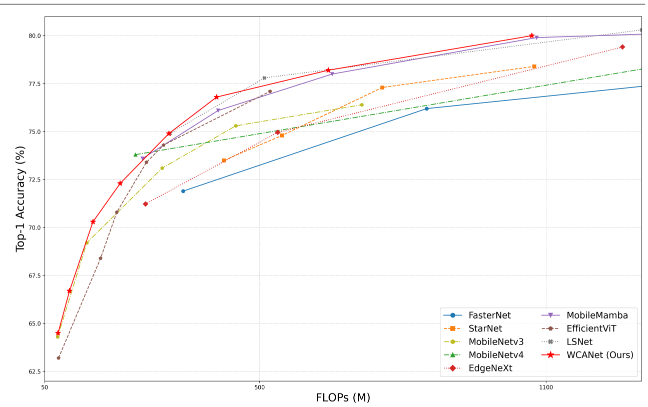

Figure 1: Performance vs. FLOPs with SoTA methods

computational redundancy but overlooked actual runtime efficiency. Some methods [27; 28] like FasterNet [29] improve the computational efficiency by reducing the extraction of redundant features. However, existing methods [27; 29] often struggle to balance representational capacity and computational efficiency in lightweight CNN design~~have not truly achieved the separation of effective features and redundant features~~. Aggressively discarding spatial details can lead to the loss of potentially useful information~~Many potentially useful features in the unutilized parts have not been extracted~~, thereby limiting the model's feature representation ability. Coupled with the limited receptive field of convolution, this has caused CNNs to lag ~~significantly~~ in accuracy compared to other architectures. To break through the performance bottleneck of CNN-based methods, this paper proposes a new lightweight feature modeling paradigm ~~feature separation method and a new feature extraction method~~.

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To achieve highly efficient spatial feature extraction ~~effective separation of redundant features~~, this paper introduces the wavelet transform [30]. We decompose the input feature map into four sub-bands, based on the differences in the information contained in each sub-band. As shown in Fig.2, the LL, LH, and HL sub-bands possess more abundant features. Therefore, we regard them as informative ~~effective~~ feature sub-bands. However, the HH sub-band only contains a small amount of high-frequency noise or sparse details~~features~~. Considering operational efficiency, we do not perform feature extraction on the HH sub-band , thereby strictly bounding the computational overhead.

Based on the above frequency-aware principles~~feature separation methods~~, this paper proposes the Cascade Channel–Spatial Attention (CCSA) module. CCSA decouples feature modeling into channel-wise and spatial levels. Firstly, CCSA uses the Effective Squeeze and Extraction (ESE) module [31] to enhance the responses of key channels; subsequently, we introduce the Learnable Wavelet Spatial Attention (LWSA) to achieve hardware-friendly lightweight CNN design and efficient spatial modeling~~the separation of redundant features in the spatial level and effective spatial modeling~~. LWSA uses the wavelet transform to separate the ~~redundant~~ HH sub-band. Then, based on the complementary characteristics of the LH and HL sub-bands, LWSA generates a spatial attention weight map, which is applied to the LL sub-band with the richest semantic information to enhance the important spatial regions. Finally, the feature map is restored to its original size through inverse wavelet transform.

Based on CCSA, this paper proposes WCANet (Wavelet Cascade Attention Network), a lightweight vision network for resource-constrained scenarios. WCANet achieves an optimal balance between accuracy and inference latency, providing a highly effective lightweight CNN design~~truly achieves the separation of effective features and redundant features, completing efficient feature modeling with extremely low computational cost~~. Fig.1 and Fig.8 show that WCANet demonstrates highly competitive performance against~~surpasses~~ existing SoTA methods. Specifically, WCANet maintains a consistent performance upper hand across the entire evaluation spectrum (from 80M to 1078M FLOPs). Experiments show that WCANet is highly effective for~~suitable for extremely~~ lightweight scenarios: on ImageNet-1K, WCANet-T1 achieves a Top-1 accuracy of 64.5% with only 80M FLOPs, surpassing EfficientViT-M0 by 1.3%. In lightweight application scenarios with more relaxed computational resources, WCANet also exhibits strong scalability: the Top-1 accuracy of WCANet-T4, T6, and T8 is 3.0, 2.0, and 1.1 percentage points higher than that of FasterNet-T0, T1, and T2, respectively. Moreover, at a comparable accuracy level, WCANet-T5 achieves approximately 42% higher GPU throughput than FasterNet-T1. Under the same throughput, WCANet achieves a performance comparable to MobileMamba while surpassing it in Top-1 accuracy across the board, and has a significantly lower inference latency on CPUs than MobileMamba.

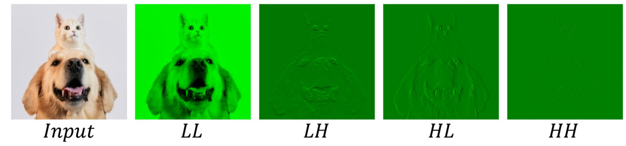

Figure 2: Wavelet decomposition sub-bands. The four sub-bands (LL, LH, HL, HH) are obtained by applying the learnable wavelet transform to an input image.

To summarize, our contributions are as follows:

- To overcome the limitation that fixed wavelet filters cannot effectively decompose high-dimensional feature maps, we propose learnable wavelet filters with orthogonality constraints, allowing filters to adapt dynamically while strictly maintaining orthogonality.
- We propose a learnable wavelet spatial attention mechanism (LWSA). LWSA leverages the features from sub-bands with different orientations to perform spatial attention on informative sub-bands, thereby enhancing key spatial nodes and achieving efficient spatial feature modeling.
- We propose Cascade Channel–Spatial Attention (CCSA). CCSA utilizes ESE and LWSA to perceive key features at both channel and spatial levels, respectively, achieving an optimized accuracy-efficiency trade-off.
- We propose WCANet, a visual network design optimized for practical lightweight applications. Extensive experiments demonstrate that WCANet exhibits robust scalability and highly competitive performance across various devices.

2. Related Work

2.1. Lightweight network design

ViT-based methods. Since the Transformer [32] was introduced to computer vision [3], various networks based on the Transformer architecture have demonstrated powerful capabilities in various computer vision tasks [33; 34; 5]. Some works have attempted to combine the ideas of ViT and CNN for application in lightweight tasks [7; 35; 36; 37]. For instance, MobileViT [10] integrates MHSA with MobileNets to achieve an architectural hybrid. EdgeViT [38] proposes the integration of self-attention and convolutions to achieve cost-effective information exchange. In addition, to enhance the actual operational efficiency, MobileViTv2 [35] proposes a linear complexity Transformer architecture to reduce the inference cost. FastViT [36] introduces structural re-parameterization to increase the running speed. SHViT[12] proposes single-head self-attention (SHSA), which improves the operational efficiency by performing attention calculations only on a portion of the channels.

SSM-based methods. State Space Models (SSMs) [14; 16; 17] have attracted considerable interest because of their computational efficiency in modeling long-range dependencies. Mamba [15] incorporates the S6 module, delivering a straightforward architecture that excels in efficiently modeling long sequences. Due to this advantage, many works have applied it to computer vision [39; 40; 18; 41]. Vision Mamba (Vim) [41] adopts bidirectional Mamba block modeling for spatial feature representation. Vmamba [18] introduces a Cross-Scan mechanism to enhance spatial modeling capabilities. EfficientVMamba [19] improves efficiency by using skip sampling. MobileMamba [20] designs a three-stage lightweight architecture with multi-receptive field interaction for high throughput.

CNN-based methods. Since AlexNet [42], CNNs have been widely applied to various computer vision tasks [43; 44; 45; 46; 47; 48]. DWConv [49] has greatly promoted the lightweight development of CNN. MobileNets [23; 21; 11], ShuffleNets [26; 25], and EfficientNet [24] reduce computational redundancy by introducing DWConv and combine ingenious network structure designs to achieve powerful feature extraction capabilities at low computational costs. GhostNet [27] further observes that there is a high degree of redundancy among the channels of feature maps. By performing cheap linear operations only on some channels, it achieves efficient feature extraction capabilities. FasterNet [29] achieves a significant improvement in operational efficiency by using PConv instead of DWConv.

2.2. Wavelet Transforms in Deep Learning

Wavelet transform [30] was introduced to enhance the performance of the model in aspects such as multi-scale feature extraction, computational efficiency, or generation quality. The wavelet transform is used as the pooling operator in convolutional networks [50; 51]. The wavelet transform is also employed to compress feature maps to reduce the computational cost of CNNs [52]. In terms of network architecture design, some researchers have utilized wavelets for downsampling in the U-Net framework [53] and implemented upsampling through inverse wavelet transform [54; 55]. The WTConv [56] combines wavelet decomposition with convolution operations, effectively expanding the receptive field and enhancing the response to low-frequency structures. However, while wavelet transforms have been successfully integrated into CNNs, current approaches typically process both high-frequency and low-frequency sub-bands equally without distinguishing their semantic richness. Further exploration is still needed to leverage the complementary characteristics of these sub-bands for dynamic feature modulation, especially under strictly limited computational budgets.

3. Method

3.1. Revisiting Convolutions

The process of the standard convolution operation is as follows: given the input feature map $X \in \mathbb{R}^{C\times H\times W}$ and the Standard Kernel $K \in \mathbb{R}^{C_{out}\times C_{in}\times H\times W}$, the output feature map $Y \in \mathbb{R}^{C_{out}\times H\times W}$ is:

$$Y = X * K, \tag{1}$$

Specifically, $y_{[c',i,j]}$ is an element in the location (i, j) at channel c' of Y. The value of $y_{[c',i,j]}$ is calculated by:

$$y_{[c',i,j]} = \sum_{c=1}^{C_{in}} \sum_{m=1}^{K} \sum_{n=1}^{K} x_{[c,i+m,j+n]} \cdot k_{[c',c,m,n]} \tag{2}$$

As shown in Eq.2, standard convolution performs local spatial feature extraction and channel-wise information fusion in one step [23]. Practice has proved that this way of feature extraction is effective [57; 58]. However, its computation complexity is relatively high$\big(O(HWK^2C_{in}C_{out})\big)$. This makes it difficult to satisfy the lightweight requirements.

The depthwise separable convolution splits the operation of Eq.2 into two layers: a depthwise convolution (DWConv) for filtering and a pointwise convolution (PWConv) layer for combining. The calculations for the two separate layers are respectively shown in Eq.3 and Eq.4.

$$\hat{x}_{c,i,j} = \sum_{m=1}^{K} \sum_{n=1}^{K} x_{c,i+m,j+n} \cdot d_{c,m,n} \tag{3}$$

$$Y = \hat{X} * Conv_{1\times 1}. \tag{4}$$

This decomposition drastically reduces the computation from $(O(HWK^2C_{in}C_{out})$ to $O(HWK^2C_{in}+O(HWC_{in}C_{out})))$.

Partial convolution (Pconv) is proposed to reduce computational redundancy and memory access simultaneously. It simply applies a regular Conv on only a part of the input channels for spatial feature extraction, and leaves the remaining channels untouched.

$$Y = [X_{1:r} * K_{1:r}, X_{r+1:C_{in}}]. \tag{5}$$

The computation complexity of PConv is $O(HWK^2C_{r\cdot in}C_{r\cdot out})$. Under the same FLOPs, PConv runs faster than DWConv [29].

Although DWConv significantly reduces the computational load, its spatial modeling capability is relatively weak. To make up for the performance loss, in practice, the number of channels of DWConv often needs to be greatly increased, which instead affects the computational efficiency. PConv improves the operational efficiency by only performing convolution operations on a small portion of the feature channels. However, a large amount of potentially useful information in the remaining channels is not extracted, which instead limits the model's feature expression ability.

To achieve truly powerful and efficient feature modeling capabilities under low computational cost, we propose Cascade Channel-Spatial Attention (CCSA). CCSA enables the network to pay more attention to key features when modeling feature maps by cascading channel attention and

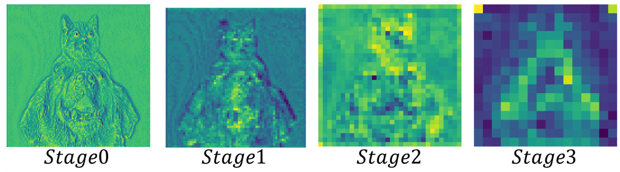

Figure 3: Feature map evolution across network stages. Visualization of feature maps at different stages (Stage 0 to Stage 3).

spatial attention. CCSA reduces computational complexity by separately modeling space and channel. By cascading channel attention and spatial attention, the network pays more attention to key features when modeling feature maps.

3.2. Learnable Wavelet Spatial Attention (LWSA)

To achieve a more powerful spatial feature modeling capability, we propose Learnable Wavelet Spatial Attention. By using the wavelet transform [30], the feature map is converted into feature sub-bands containing different information. To enhance computational efficiency, LWSA only performs feature extraction on sub-bands that contain more information. To improve the modeling ability of key spatial features, we take advantage of the characteristic that sub-bands contain different information and construct Wavelet Spatial Attention (WSA).

Previous studies [56; 59; 20] often relied on predefined fixed filters when conducting wavelet transforms, such as Haar Wavelet [52; 60; 61]. After wavelet transformation, the input feature map is transformed into four sub-bands containing different information. Next, existing methods, such as WTConv [56], often extract features from all sub-bands by directly applying standard convolution operations.However, this paradigm introduces two major limitations. First, high-level feature maps contain rich semantic information, as shown in Fig.3. Fixed filters cannot effectively separate the information of high-level feature maps. Second, existing convolution-based feature extraction methods cannot achieve a powerful spatial feature modeling capability. To solve these problems, we innovatively propose learnable wavelet spatial attention (LWSA). It consists of the following two components.

To overcome the first problem, we propose the Learnable Wavelet Basis. Furthermore, we construct **Learnable Wavelet Filters (LWF)** to replace the fixed filters. This not only satisfies orthogonality but also enhances the filter's adaptability. Given an input feature map $X \in \mathbb{R}^{C\times H\times W}$, network generate a two-dimensional wavelet filters group $F \in \mathbb{R}^{4\times2\times2}$ for each channel slice $x \in \mathbb{R}^{1\times H\times W}$. Specifically, we first initialize c one-dimensional learnable low-pass filters $g_{lo} \in \mathbb{R}^2$. The corresponding c high-pass filters g_{hi} are explicitly constructed according to the orthogonality condition:

$$g_{lo}^{\text{init}} = \frac{1}{\sqrt{2}}\begin{bmatrix}1 & 1\end{bmatrix}, g_{hi} = \begin{bmatrix}g_{lo}[1] & -g_{lo}[0]\end{bmatrix}. \tag{6}$$

Building upon this, we construct a two-dimensional analysis filter group F through the matrix multiplication operation:

$$\begin{aligned} F &= \begin{bmatrix}F_{\text{LL}} F_{\text{LH}} F_{\text{HL}} F_{\text{HH}}\end{bmatrix} \\ F_{\text{LL}} = g_{\text{lo}} \otimes g_{\text{lo}}, &\quad F_{\text{LH}} = g_{\text{lo}} \otimes g_{\text{hi}}, \\ F_{\text{HL}} = g_{\text{hi}} \otimes g_{\text{lo}}, &\quad F_{\text{HH}} = g_{\text{hi}} \otimes g_{\text{hi}}. \end{aligned} \tag{7}$$

To efficiently implement multi-channel wavelet decomposition, we integrate the c two-dimensional filter groups into a single depthwise separable convolution kernel $W^{DW} \in \mathbb{R}^{4c\times1\times2\times2}$. During the backward pass, g_{lo} is updated by gradient descent. Meanwhile, g_{hi} and W^{DW} are dynamically generated from g_{lo} through orthogonality constraints and tensor operations , which avoids the gradient instability and divergence issues caused by non-orthogonality. A similar hard-constraint design is also reflected in other domains to ensure optimization stability and theoretical validity [62].

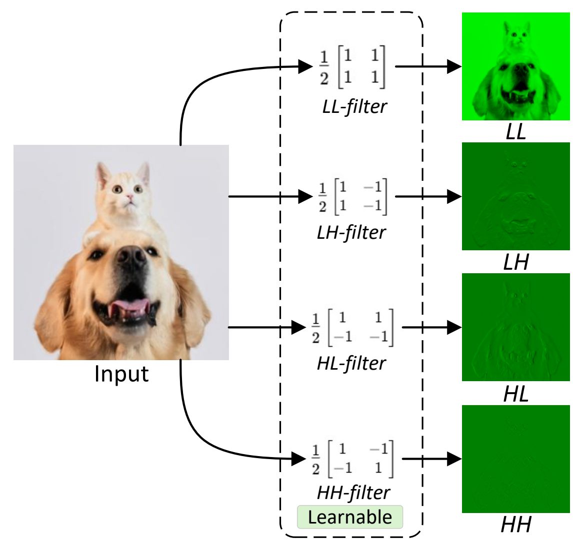

Figure 4: The process of Wavelet Transform.

To fix the second problem, we propose **Wavelet Spatial Attention** to replace sample concatenation and convolutions. As shown in Fig.4, the LL sub-band retains the most details of the input feature map. The LH sub-band and HL sub-band highlight the vertical and horizontal information, respectively. Therefore, we enhance the information of the LL sub-band by utilizing the information of the LH and HL sub-bands. Fig.5 (c) shows the specific operation. We first perform local convolutional feature extraction on HL and LH by leveraging DWConv, and then perform the Hadamard product [1] on the two obtained feature maps. Finally, the HardSigmoid activation function is used to generate the spatial attention matrix A.

$$\begin{aligned} \tilde{\text{LH}} &= \text{DWConv}_{3\times3}(\text{LH}) \\ \tilde{\text{HL}} &= \text{DWConv}_{3\times3}(\text{HL}) \\ A &= \text{Hardsigmoid}\left(\tilde{\text{LH}} \odot \tilde{\text{HL}}\right) \end{aligned} \tag{8}$$

[1]Here, the symbol $\odot$ strictly denotes element-wise multiplication.

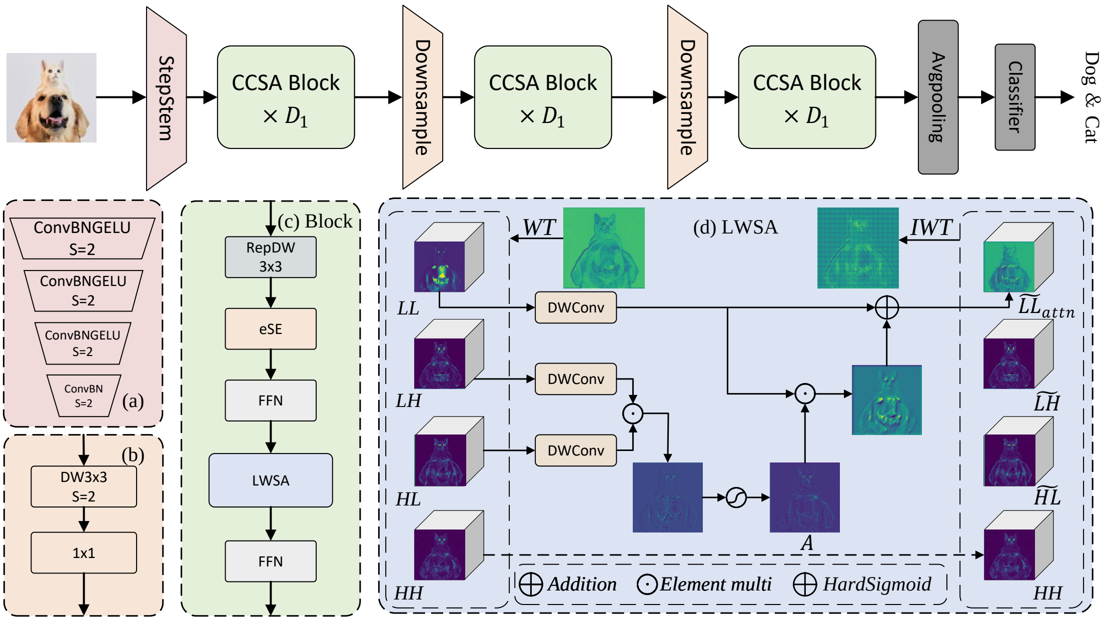

Figure 5: The structure of WCANet. (a) is the StepStem. (b) is the CCSA Block. (c) is the LWSA.

where the shape of $A \in \mathbb{R}^{B \times C \times H/2 \times W/2}$ is the same as that of the LL sub-band. Thus, spatial attention is achieved through the operation as shown in the formula X:

$$\tilde{\mathrm{LL}}_{\mathrm{attn}} = A \odot \mathrm{DWConv}_{k \times k}(\mathrm{LL}) + \mathrm{LL}, \tag{9}$$

where k represents the size of the convolution kernel. Its value can be dynamically set according to the stage of the network $\left(Stage0 : (k = 7), Stage1 : (k = 5), Stage \geq 2 : (k = 3)\right)$.

$$IWT(\tilde{\mathrm{LL}}_{\mathrm{attn}}, \tilde{\mathrm{LH}}, \tilde{\mathrm{HL}}, HH) = X^{\mathrm{out}} \tag{10}$$

We concatenate $\tilde{\mathrm{LL}}_{\mathrm{attn}}$,$\tilde{\mathrm{LH}}$,$\tilde{\mathrm{HL}}$ with the original HH along the channel-wise, and then perform an inverse wavelet transform to restore the feature map to its original size.

3.3. The design of WCANet

We present the lightweight model design, i.e., WCANet, as shown in Fig.5. It uses a sequence attention as the primary operation. We present the basic block, i.e., CCSA block in Fig.5 (b).

CCSA block applies attention operations separately along the channel and spatial dimensions, which achieves a powerful feature extraction capability. In the channel dimension, the ESE module is employed to suppress redundant channels and enhance effective channels. In the spatial dimension, we employ LWSA to achieve effective spatial feature extraction. Besides, we utilize the extra depthwise convolution and FFN to enhance the model's capability.

We compare the ×32 and ×64 downsampling multi-stage architectures. As shown in Tab 1, under different network architectures, ×64 downsampling is more efficient than ×32 downsampling. Furthermore, the experimental results show that WCANet-T4×64 achieves a Top-1 accuracy 0.8 ↑ than its ×32 counterpart, and WCANet-T5×64 matches the Top-1 accuracy of the ×32 version. Therefore, we utilize the 64× downsampling in WCANet.

Overall, WCANet employs a StepStem [63] to project the input image into the visual feature map. For downsampling, we leverage the depth-wise and point-wise convolution to reduce the spatial resolution and modulate the channel dimension, respectively. Besides, we stack CCSA blocks in all stages. Finally, the feature map is processed by global average pooling and a classifier to produce the final class prediction.

Table 1
Comparison of WCANet architectures with different total downsampling ratios: WCANet-T4×32 and WCANet-T5x32 denote 4-stage networks with a total downsampling factor of 32, while WCANet-T4×64 and WCANet-T5×64 denote 3-stage networks with a total downsampling factor of 64.

model	network structure	Throughput (images/s)	FLOPs (M)
WCANet-T4 ×32	$\{32, 64, 128, 256\}$ $\{1, 2, 4, 5\}$	5334	275
WCANet-T4 ×64	$\{128, 256, 512\}$ $\{2, 3, 2\}$	9921	313
WCANet-T5 ×32	$\{40, 80, 160, 320\}$ $\{1, 2, 4, 5\}$	4112	442
WCANet-T5 ×64	$\{128, 256, 512\}$ $\{3, 4, 3\}$	7472	410

Table 2
Classification results of different models on ImageNet-1K.

Model	Params (M)	FLOPs (M)	Reso.	Top-1 (%)	#Pub
EfficientViT-M0 [9]	2.3	79	224	63.2	CVPR'23
MobileNetv3-Lx035 [64]	2.1	77	224	64.2	ICCV'19
WCANet-T1	2.0	78	256	64.5	-
WCANet-T2	2.5	102	256	**66.7**	-
EfficientViT-M1 [9]	3.0	167	224	68.4	CVPR'23
MobileNetv3Lx05 [64]	2.6	138	224	68.8	ICCV'19
ShufflenetV2-1.0x [25]	2.3	146	224	69.4	ECCV'18
EfficientViT-M2 [9]	4.2	201	224	70.8	CVPR'23
WCANet-T2d5	3.7	151	256	70.3	-
WCANet-T3	5.1	208	256	**72.3**	-
EdgeNeXt-XXS [8]	1.3	260	224	71.2	ECCV'22
FasterNet-T0 [29]	3.9	340	224	71.9	CVPR'23
ShuffleNetV2-1.5x [25]	3.5	300	224	72.6	ECCV'18
EMO-1M [65]	1.3	261	224	71.5	ICCV'23
SHViT-S1 [12]	6.3	241	224	72.8	CVPR'24
EfficientViT-M3 [9]	6.9	263	224	73.4	CVPR'23
StarNet-S1 [66]	2.9	425	224	73.5	CVPR'24
MobileMamba-T2 [20]	8.8	255	192	73.6	CVPR'25
EfficientViT-M4 [9]	8.8	299	224	74.3	CVPR'23
LSNet-T [67]	11.4	320	224	74.9	CVPR'25
EMO-2M [65]	2.3	439	224	75.1	ICCV'23
SHViT-S2 [12]	11.4	366	224	75.2	CVPR'24
MobileMamba-T4 [20]	14.2	413	192	76.1	CVPR'25
WCANet-T4	7.6	313	256	74.9	-
WCANet-T5	10.8	410	256	**76.8**	-
StarNet-S2 [66]	3.7	547	224	74.7	CVPR'24
EdgeNeXt-XS [8]	2.3	540	224	75.0	ECCV'22
FasterNet-T1 [29]	7.6	850	224	76.2	CVPR'23
EfficientViT-M5 [9]	12.4	522	224	77.1	CVPR'23
SHViT-S3 [12]	14.2	601	224	77.4	CVPR'24
MobileMamba-S6 [20]	15.0	652	224	78.0	CVPR'25
WCANet-T6	16.5	644	256	**78.2**	-
StarNet-S3 [66]	5.8	757	224	77.4	CVPR'24
RepViT-M0.9 [68]	5.1	846	224	77.4	CVPR'24
EdgeViT-XS [38]	6.7	1100	224	77.5	ICCV'22
StarNet-S4 [66]	7.5	1075	224	78.8	CVPR'24
FasterNet-T2 [29]	15.0	1917	224	78.9	CVPR'23
EMO-6M [65]	6.1	961	224	79.0	ICCV'23
SHViT-S4 [12]	16.5	986	256	79.4	CVPR'24
EfficientViT-M4 [9]	12.4	1486	384	79.8	CVPR'23
MobileMamba-B1 [20]	17.1	1080	256	79.9	CVPR'25
WCANet-T8	26.0	1078	256	**80.0**	-

We build multiple WCANet variants to accommodate different computational budgets. Under an input resolution of 256×256, their FLOPs range from 80M to 650M, namely WCANet-T0, WCANet-T1, WCANet-T2, WCANet-T2P5, WCANet-T3, WCANet-T4, WCANet-T5, and WCANet-T6. Additionally, to better suit downstream dense prediction tasks, we also design a four-stage WCANet variant with 64× downsampling: WCANet-T8.

4. Experiments

4.1. Image Classification

We conduct experiments on the ImageNet-1K [69] dataset to evaluate the performance of WCANet on the image classification task. To ensure a fair and rigorous comparison against baselines spanning different publication years, we explicitly document our training protocol. Specifically, WCANet is trained entirely from scratch without relying on Knowledge Distillation (KD), additional pre-training datasets, or multi-scale training procedures. The detailed hyperparameters and data augmentation strategies are fully documented in Table 12. The overall training protocol is similar to that of [9; 67; 20], except that WCANet uses an input image resolution of 256 × 256 and a learning rate that is cosine-annealed from 3e-3 to 1e-5. As WCANet incorporates wavelet transform mechanisms across multiple stages, the maximum downsampling factor of the network reaches 128. To ensure the complete alignment of feature dimensions, the input resolution must be perfectly divisible by this factor. Therefore, WCANet uses an input image resolution of 256 × 256.

Table 2 shows the performance of WCANet, compared with other state-of-the-art (SoTA) methods under similar model scales. The different model scales are categorized by FLOPs. Compared with CNNs, WCANet demonstrates consistently superior performance at the same computational cost. Specifically, WCANet-T4 outperforms FasterNet-T0 [29] by +3 ↑. WCANet-T8 surpasses FasterNet-T2 [29] by 1.1 ↑ in Top-1 while having only 56% of its FLOPs. Furthermore, WCANet variants consistently surpass StarNet [66] counterparts by 1.2 ↑ to 3.3 ↑ in Top-1. Compared with ViTs, ViMs, and hybrid architectures, WCANet also demonstrates strong competitiveness. Under comparable FLOPs,

WCANet variants consistently outperform the Transformer-based EfficientViT [9] series in Top-1 accuracy. WCANet-T4 achieves 74.9% Top-1 accuracy, surpassing the Mamba-based MobileMamba-T2 [20] by +1.3 ↑. At approximately 1 GFLOPs, WCANet-T8 achieves 80.0% Top-1 accuracy, outperforming contemporary architectures at similar computational costs, including MobileMamba-B1 [20], EMO-6M [65], and ShViT-S4 [12]. These experimental results strongly indicate that by employing LWSA for spatial feature extraction and the ESE module for inter-channel information fusion, the proposed model performs very well in terms of classification accuracy.

4.2. Inference Efficiency Analysis

To evaluate the hardware-aware inference efficiency of WCANet, we conduct throughput benchmarks on three representative hardware platforms: an NVIDIA GeForce RTX 4090 GPU, an Intel(R) Xeon(R) Platinum 8352V CPU, and an Apple M1 (ARM-based chip). The batch sizes are set to 2048, 16, and 8 for the GPU, CPU, and Apple M1 platforms, respectively. Each model is evaluated at the input resolution listed in Table 3. For each model-device pair, the throughput

Figure 6: Qualitative results for object detection and instance segmentation on COCO-2017 using WCANet as the backbone framework.

Table 3
Comparison with SoTAs in inference efficiency. GPU throughput is measured on an NVIDIA GeForce RTX 4090 with a batch size of 2048; CPU throughput is evaluated on an Intel(R) Xeon(R) Platinum 8352V with a batch size of 16; and ARM throughput is evaluated on an Apple M1 chip with a batch size of 8. Each model is evaluated at the input resolution listed in the "Reso." column. Throughput is reported as mean ± standard deviation over five repeated runs.

Model	FLOPs (M)	#Params (M)	Reso.	Throughput (img/s) GPU	CPU	ARM	Top-1 %
StarNet-S1	425	2.9	224	6898±55	42.7±0.8	25.8±0.5	73.5
FasterNet-T0	340	3.9	224	15682±141	77.4±1.2	62.6±1.0	71.9
MobileMamba-T2	255	8.8	192	19541±127	54.6±1.0	51.7±0.9	73.6
LSNet-T	320	11.4	224	16106±134	-	-	74.9
WCANet-T4	313	7.6	256	20274±146	60.6±1.1	48.7±0.8	74.9
FasterNet-T1	850	7.6	224	10731±163	38.7±0.7	42.6±0.8	76.2
MobileMamba-T4	413	14.2	192	15418±128	40.7±0.8	38.8±0.7	76.1
WCANet-T5	410	10.8	256	15241±143	41.7±1.4	34.8±0.6	76.8
FasterNet-T2	1917	15.0	224	4946±95	18.8±0.8	27.7±0.6	78.9
StarNet-S4	1075	7.5	224	3492±43	23.8±0.5	11.9±0.4	78.8
MobileMamba-B1	1080	17.1	256	6152±52	12.8±0.4	15.8±0.7	79.9
WCANet-T8	1078	26.0	256	6236±57	17.8±0.4	16.8±0.3	80.0

measurement is repeated five times, and the results are reported as mean ± standard deviation. All models are benchmarked in PyTorch using FP32 precision, and all operators are implemented with official library components without additional model-specific deployment optimizations.

As summarized in Table 3, WCANet achieves a favorable accuracy–efficiency trade-off across different hardware platforms. On the RTX 4090, WCANet-T5 achieves a throughput of 15,241 ± 143 img/s with 76.8% Top-1 accuracy, compared with 10,731±163 img/s and 76.2% Top-1 accuracy for FasterNet-T1. Thus, at a comparable accuracy level, WCANet-T5 provides approximately 42% higher throughput. In addition, WCANet-T4 achieves 20,274 ± 146 img/s, which is approximately 2.9× the throughput of StarNet-S1 (6,898 ± 55 img/s), while improving Top-1 accuracy by 1.4 percentage points.

The performance trade-offs are further visualized in Figure 8, which plots the relationship between inference latency/throughput and Top-1 accuracy across different hardware platforms. As illustrated, WCANet consistently lies near the favorable frontier of the accuracy–efficiency trade-off compared with FasterNet, StarNet, and MobileMamba. These results indicate that WCANet effectively balances feature modeling capability and computational overhead, making it suitable for resource-constrained applications where both predictive accuracy and hardware-specific inference efficiency are important.

4.3. Object Detection and Segmentation on COCO

To further evaluate the generalization ability of WCANet, we conduct object detection and instance segmentation experiments on the COCO dataset [72]. We first pre-train the WCANet in ImageNet-1k, and then use it as the backbone integrated with both RetinaNet [73] and Mask R-CNN [74] frameworks. The models are trained for 12 epochs (1× schedule) with a total batch size of 16, using an AdamW optimizer [75] (learning rate 1×10^{-4}, weight decay 0.05). The learning rate is linearly warmed up over the first 500 iterations and then decayed via cosine annealing, with gradient clipping (max norm 0.1). No additional hyper-parameter tuning is performed.

Table 4 shows the comparison between WCANet and representative models. WCANet consistently achieves higher average precision (AP) than conventional CNN baselines under comparable model complexity. Specifically, WCANet-T8 outperforms the standard ResNet50 [2] by +2.3 AP^b with RetinaNet, and by +2.0 AP^b and +2.3 AP^m with Mask R-CNN. Moreover, WCANet-T8 shows competitive performance against the efficient CNN architecture FasterNet-S [29], achieving +0.1 higher AP^b (40.0 vs.39.9) but slightly lower AP^m (36.7 vs.36.9) under Mask R-CNN.

Table 4
Object detection and instance segmentation results on COCO. AP^b and AP^m indicate bounding box AP and mask AP, respectively. For a fair comparison, all baselines are evaluated according to the training protocols reported in their original publications.

Backbone	RetinaNet						Mask R-CNN					
	AP	AP_{50}	AP_{75}	AP_S	AP_M	AP_L	AP^b	AP^b_{50}	AP^b_{75}	AP^m	AP^m_{50}	AP^m_{75}
EfficientViT-M5 [9]	34.3	54.2	36.1	18.0	36.9	48.2	34.9	57.0	37.0	32.8	53.7	34.6
SHViT-S3 [12]	36.1	56.6	38.0	19.9	39.1	50.8	36.9	59.4	39.6	34.4	56.3	36.1
ResNet50 [2]	36.3	55.3	38.6	19.3	40.0	48.8	38.0	58.6	41.4	34.4	55.1	36.7
PVT-Tiny [70]	36.7	56.9	38.9	**22.6**	38.8	50.0	36.7	59.2	39.3	35.1	56.7	37.3
PoolFormer-S12 [71]	36.2	56.2	38.2	20.8	39.1	48.0	37.3	59.0	40.1	34.6	55.8	36.9
LSNet-S [67]	36.7	67.2	38.6	20.0	39.7	51.8	37.4	59.9	39.8	34.8	56.8	36.6
FasterNet-S [29]	-	-	-	-	-	-	39.9	61.2	**43.6**	**36.9**	58.1	**39.7**
WCANet-T8	**38.6**	**59.0**	**41.1**	22.5	**41.5**	**51.7**	**40.0**	**61.3**	43.4	36.7	**58.6**	39.1

Table 5
Performance comparison on the ClinicDB dataset over five independent data splits. For each split, 80% of the images are used for training, 10% for validation, and 10% for testing. For each data split, all compared models use the same training, validation, and testing subsets and the same model-training seed (2026). The results are reported as mean ± standard deviation across the five data splits.

Method	**Val Dice (%)**	**Val IoU (%)**	**Test Dice (%)**	**Test IoU (%)**
MK_UNet	89.91 ± 0.63	83.46 ± 0.93	93.59 ± 0.98	88.46 ± 1.43
WCANet (Fixed Haar Filters)	90.95 ± 0.39	84.91 ± 0.56	93.75 ± 0.67	88.77 ± 0.72
WCANet (Learnable Filters, Ours)	**91.16 ± 0.60**	**85.43 ± 0.87**	**94.28 ± 0.60**	**89.26 ± 0.75**

To provide a qualitative evaluation of the proposed network, we visualize the object detection and instance segmentation results in Fig. 6. Notably, WCANet exhibits robust detection and precise segmentation even for small and texture-rich objects. While the LWSA module discards the HH sub-band to optimize computational efficiency, the visualization and our quantitative results (e.g., 22.5% AP_S in RetinaNet) demonstrate that the remaining LH and HL sub-bands sufficiently capture the critical high-frequency spatial details, such as edges and boundaries, required for dense prediction tasks. Therefore, WCANet successfully maintains high sensitivity to small objects while strictly bounding the computational overhead.

4.4. Medical Image Segmentation on ClinicDB

To evaluate how WCANet behaves across domains where the frequency characteristics of meaningful features differ substantially from natural ImageNet statistics, we conduct further experiments on medical image segmentation using the ClinicDB dataset. Medical images often exhibit highly unique texture, contrast, and edge distributions compared to natural scenarios. Given the relatively small size of ClinicDB, performance obtained from a single data partition may be sensitive to the specific training, validation, and testing split. Therefore, we further evaluate all compared models over five independent data splits. For each split, 80% of the images are used for training, 10% for validation, and 10% for testing. For each split, all compared models use exactly the same training, validation, and testing subsets, while the model-training seed is fixed to 2026. The results are reported as mean ± standard deviation across the five independent splits.

As shown in Table 5, MK_UNet [76] achieves 93.59 ± 0.98% Test Dice and 88.46 ± 1.43% Test IoU. Introducing fixed Haar filters into WCANet yields 93.75 ± 0.67% Test Dice and 88.77 ± 0.72% Test IoU. In contrast, WCANet equipped with our proposed learnable wavelet filters achieves the best average performance, reaching 94.28 ± 0.60% Test Dice and 89.26 ± 0.75% Test IoU. Compared with the fixed-Haar variant, the learnable filters improve the average Test Dice and Test IoU by 0.53 and 0.49 percentage points, respectively. Meanwhile, the standard deviations of the learnable-filter variant remain comparable to those of the fixed-Haar counterpart across different data partitions, indicating that the observed performance improvement is robust to variations in the dataset split rather than being specific to a single partition.

These results further demonstrate the cross-domain adaptability of the proposed learnable wavelet filters. The learnable decomposition can adjust to the distinctive texture, contrast, and boundary characteristics of medical images. Meanwhile, the orthogonality constraint maintains a mathematically structured and non-redundant multi-band decomposition during optimization.

To intuitively analyze the effectiveness of our proposed WCANet in spatial feature extraction, we visualize the intermediate feature responses of MK-UNet, WCANet with fixed Haar filters, and WCANet with learnable filters in the

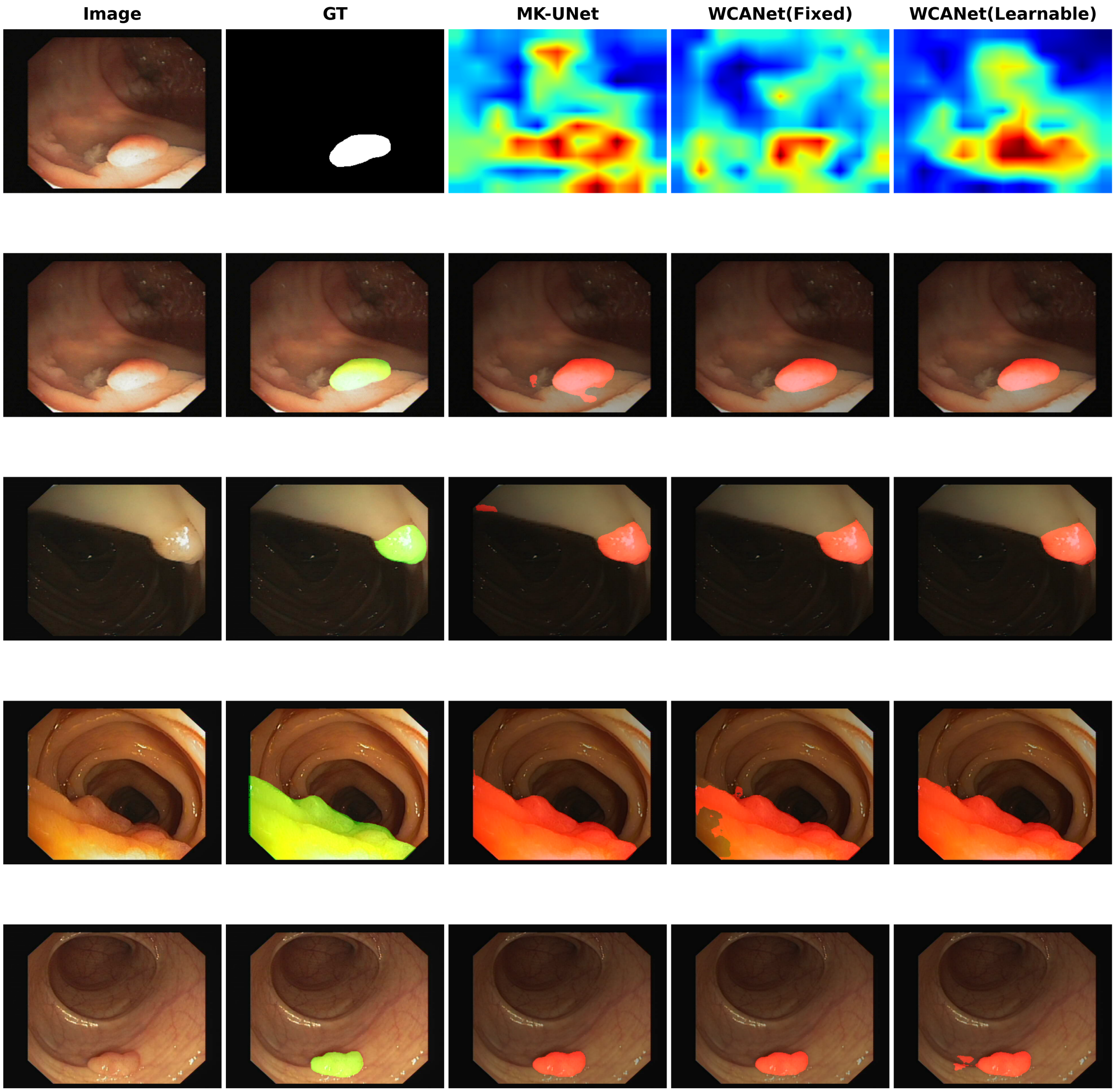

Figure 7: Qualitative feature-response and segmentation comparison on the ClinicDB dataset. The first row visualizes the intermediate spatial responses of MK-UNet, WCANet with fixed Haar filters, and WCANet with learnable filters, with the input image and ground truth provided as references. The following four rows present qualitative segmentation results. In the representative cases, the learnable-filter variant produces more target-focused feature responses and cleaner segmentation masks. The last row additionally presents a challenging example in which the improvement over the fixed-filter variant is less pronounced.

first row of Fig. 7, together with representative segmentation results in the subsequent rows. Compared with MK-UNet, the wavelet-based variants suppress more irrelevant background responses, while the learnable-filter variant further produces a more compact and target-focused activation pattern. This observation suggests that adaptive frequency decomposition can better preserve task-relevant information and suppress less informative responses, which is consistent with the general feature-engineering principle of improving the discriminative quality of learned representations [77]. The segmentation examples further show that WCANet with learnable filters generally produces cleaner target regions and more accurate localization than the baseline and fixed-filter variants, supporting the effectiveness of the proposed Learnable Wavelet Spatial Attention for spatial feature modeling.

4.5. Model Analyses

In this subsection, we conduct ablation experiments by using WCANet on ImageNet-100, which is a subset of ImageNet-1K. All models are trained under the same settings [78; 9] with 100 epochs.

Analysis of Learnable Wavelet Filters. To comprehensively characterize both the individual contributions and cross-stage interactions of learnable wavelet filters, we conduct an exhaustive combinatorial ablation on WCANet-T1 covering all possible configurations across Stage 0, Stage 1, and Stage 2. As shown in Table 6, enabling learnable filters

Table 6
Combinatorial ablation study of learnable filters across different stages on WCANet-T1.

Configuration	FLOPs (M)	Top-1 Acc (%)	Δ (%)
Baseline (None)	79.55	71.6	-
Stage 0 Only	79.55	71.9	+0.3
Stage 1 Only	79.55	72.6	+1.0
Stage 2 Only	79.55	72.6	+1.0
Stage 0 & 1	79.55	72.6	+1.0
Stage 0 & 2	79.55	72.5	+0.9
Stage 1 & 2	79.55	72.7	+1.1
All Stages	79.55	72.9	+1.3

at any single stage improves performance over the fixed-filter baseline of 71.6%, while the pairwise configurations further allow us to assess whether these stage-wise gains accumulate independently. Stage 1 and Stage 2 each provide a 1.0 percentage-point improvement, substantially larger than the 0.3 percentage-point gain obtained at Stage 0, indicating that learnable filters are more beneficial in the middle and deeper stages. This may be attributed to the more abstract and semantically complex representations at later network depths, where adaptive frequency decomposition can provide greater flexibility than fixed Haar filters. However, the pairwise results show that the stage-wise gains are not strictly additive: Stage 0 & Stage 1 reaches 72.6%, while Stage 0 & Stage 2 reaches 72.5%, neither exceeding the corresponding Stage 1 or Stage 2 single-stage configuration. By contrast, jointly enabling Stage 1 and Stage 2 achieves 72.7%, slightly outperforming either stage alone and suggesting complementary effects between the middle and deeper stages despite partial overlap across stages. Finally, enabling learnable filters at all three stages achieves the best accuracy of 72.9% (+1.3 percentage points), indicating that multi-stage adaptive decomposition provides the strongest overall configuration, while the observed improvements arise from complementary but partially overlapping cross-stage effects rather than a simple additive accumulation of independent stage-wise gains.

Table 7
LWSA and others.

Strategy	FLOPs	Top-1
Identity	313	75.2
7×7 DW	318	79.1
LS Conv	321	80.4
LWSA	313	**80.6**

Analysis of Wavelet Spatial Attention. To verify the rationality of the Wavelet Spatial Attention (WSA) design proposed in LWSA, we compare LWSA with other feature extraction methods. We design three approaches: the first one directly applies an identity transformation to the LL, LH, and HL sub-bands; the second one performs 7x7 depthwise convolution on each of the three sub-bands; and the third one is our proposed WSA. As shown in Tab. 7, LWSA achieved a top-1 accuracy of up to 80.6% while having fewer FLOPs than both the 7x7 depthwise convolution and the LS Conv method. To verify whether the HH sub-band is beneficial to model performance, we conducted experiments by convolving all sub-bands and only convolving the LL, HL, and LH sub-bands. As shown in Tab.8, the result indicates that neglecting to perform convolution on the HH sub-band leads to lower FLOPs without significant changes in top-1 accuracy, validating the effectiveness of the feature separation method proposed in LWSA.

Table 8
HH sub-hand method

Strategy	FLOPs	Top-1
ALL DW	318	79.1
HH Identity	316	79.0

Table 9
All variants are based on WCANet-T4 and evaluated on ImageNet-100 with 100 training epochs.

Function	60 Epoch Top1-Acc(%)	100 Epoch Top1-Acc(%)
HL	73.01%	78.04%
HH	73.16%	77.92%
LL	71.92%	77.31%
$HL \odot LH$	77.58%	80.57%
$HH \odot LH$	76.71%	80.64%
$HL \odot HH$	76.64%	80.61%
$HL \odot LH \odot HH$	76.21%	80.54%

Analyze the effectiveness of different feature weight generation methods. We evaluated the impact on model performance when feature weights are generated from one, two, or three subbands. The experimental results are shown in Table 9. It can be observed that feature weights derived from a single subband yield the lowest Top-1 accuracy, whereas those generated from two or three subbands achieve better performance. However, compared to other approaches, the $HL \odot LH$ variant demonstrates superior convergence over variants using three or other combinations of subbands. This strongly validates the effectiveness of the $HL \odot LH$ method.

Table 10
Ablation study of individual components (ESE and LWSA) within the CCSA block.

Configuration	Top-1 Acc (%)
Baseline (None)	80.2
ESE Only	82.7
LWSA Only	83.3
CCSA (ESE + LWSA)	**85.5**

Ablation Study of CCSA Components. To isolate and quantify the individual contributions of the ESE module and the Learnable Wavelet Spatial Attention (LWSA), we conduct a component-wise ablation study on the CCSA block. As shown in Table 10, both components independently improve model performance. Compared to the baseline model (80.2%), introducing only the ESE module or only the LWSA module yields significant improvements of +2.5% and +3.1%, respectively. The slightly better independent performance of LWSA (83.3%) over ESE (82.7%) indicates that our wavelet-based spatial attention plays a particularly crucial role in visual feature extraction. Furthermore, combining ESE and LWSA into the complete cascaded CCSA block achieves the highest Top-1 accuracy (85.5%). This confirms that channel-wise and spatial-level feature enhancements are highly complementary, and the cascaded design is strictly necessary for comprehensive and effective feature extraction.

Table 11
Ablation study of filter types on COCO object detection (RetinaNet 1x schedule).

Filter Type	AP	AP_{50}	AP_{75}	AP_S	AP_M	AP_L
Fixed Haar Filters	37.8	58.2	40.2	21.3	40.9	51.0
Learnable Filters (Ours)	**38.6**	**59.0**	**41.1**	**22.5**	**41.5**	**51.7**
Degradation (Δ)	*-0.8*	*-0.8*	*-0.9*	*-1.2*	*-0.6*	*-0.7*

Impact of Learnable Filters on Downstream Tasks. *To further validate the effectiveness of our learnable wavelet filters in dense prediction, we conducted an ablation study on the COCO dataset using the RetinaNet framework by replacing the learnable filters with fixed Haar filters. As shown in Table 11, reverting to fixed filters leads to a noticeable degradation in overall performance (AP drops by 0.8%). Notably, the accuracy degradation differs across object categories. The most severe drop occurs in small objects (AP_S decreases by 1.2%), followed by large objects (-0.7%) and medium objects (-0.6%). This scale-aware degradation strongly indicates that data-driven learnable filters are particularly crucial for adaptively distinguishing fine-grained details and subtle edges from background noise, an adaptability that rigid fixed filters fail to provide.*

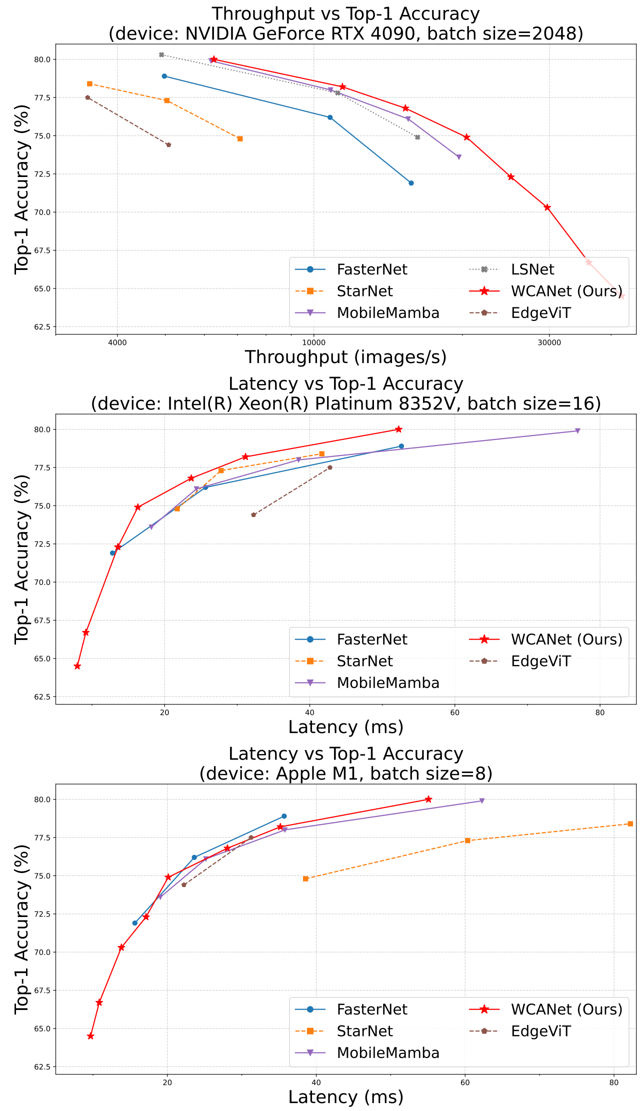

Figure 8: Accuracy–efficiency trade-offs of WCANet and representative models across different hardware platforms. GPU throughput is evaluated on an NVIDIA GeForce RTX 4090 with a batch size of 2048, while CPU and ARM latency are evaluated on an Intel(R) Xeon(R) Platinum 8352V with a batch size of 16 and an Apple M1 (ARM-based chip) with a batch size of 8, respectively. Each point is obtained from the mean of five repeated measurements, with the corresponding standard deviations reported in Table 3.

5. Conclusion

In this work, we propose WCANet, a lightweight neural network that achieves efficient feature modeling at a low computational cost. The core of WCANet is the Cascade Channel-Spatial Attention module (CCSA). CCSA decomposes feature extraction into the channel-wise and spatial levels. In the channel-wise, CCSA employs the ESE Module to augment key feature channels. In the spatial dimension, CCSA employs Learnable Wavelet Spatial Attention (LWSA) to achieve efficient and effective spatial modeling. CCSA enables WCANet to better focus on key features under low computational cost. WCANet achieves state-of-the-art speed-accuracy trade-offs on multiple devices.

A. Appendix

In this appendix, we provide more details on the experimental setup, architecture configuration, implementation of LWSA, limitations, and future work.

A.1. ImageNet-1k experimental settings

For the image classification task on ImageNet-1k, our training scheme is shown in Tab.12. Since the maximum

```python
import torch
from torch import nn
from functools import partial
from wavelet_transform import create_learnable_wavelet_filter, wavelet_transform, inverse wavelet_transform

class LWSA(nn.Module):
    def __init__(self, dim, wt_type='db1', learnable_wavelet=False,stage=0):
        super(WTAttn, self).__init__()
        self.learnable_wavelet = learnable_wavelet
        wt_filter, iwt_filter = create_learnable_wavelet_filter(dim, dim, wt_type=wt_type, type=torch.float)
        self.wt_filter = nn.Parameter(wt_filter)
        self.iwt_filter = nn.Parameter(iwt_filter)
        self.wt_function = partial(wavelet_transform, filters=self.wt_filter)
        self.iwt_function = partial(inverse_wavelet_transform, filters=self.iwt_filter)
        self.lh_conv = nn.Conv2d(dim, dim, kernel_size=3, padding=1, groups=dim)
        self.hl_conv = nn.Conv2d(dim, dim, kernel_size=3, padding=1, groups=dim)
        if stage == 0:
            self.ll_conv = nn.Conv2d(dim, dim, kernel_size=7, padding=3, groups=dim)
        elif stage == 1:
            self.ll_conv = nn.Conv2d(dim, dim, kernel_size=5, padding=2, groups=dim)
        else :
            self.ll_conv = nn.Conv2d(dim, dim, kernel_size=3, padding=1, groups=dim)
        self.act = nn.Hardsigmoid()
    def forward(self, x):
        x_wt = self.wt_function(x)
        ll, lh, hl, hh = x_wt[:, :, 0, :, :], x_wt[:, :, 1, :, :], x_wt[:, :, 2, :, :], x_wt[:, :, 3, :, :]
        lh_conv = self.lh_conv(lh)
        hl_conv = self.hl_conv(hl)
        ll_conv = self.ll_conv(ll)
        attn = (self.act((lh_conv * hl_conv )) * ll_conv )+ ll
        wt_map = torch.cat([attn.unsqueeze(2), lh_conv.unsqueeze(2), hl_conv.unsqueeze(2), hh.unsqueeze(2)], dim=2)
        output = self.iwt_function(wt_map)
        return output
```

Figure 9: The Pytorch code of LWSA. Due to space limitations, the specific implementation method of creating learnable wavelet filters is not shown.

downsampling of WCANet is $\times 128$, we use an image size of 256×256 for both training and testing. We found that T6 and T8 did not reach the highest accuracy at 300 rounds, so the training epochs for T6 and T8 are 400 epochs. We use the AdamW optimizer in combination with a cosine learning rate scheduler. The initial learning rate is set to 3×10^{-3}, and the total batch size is set to 1024. For data augmentation, we utilized methods such as Mixup, CutMix, and Random Erasing. We also use EMA to enhance the stability of training. Furthermore, to ensure a smooth startup and prevent gradient instability during the early stages of training, we consistently employ a linear warmup strategy (e.g., 20 epochs for ImageNet-1k) before the cosine learning rate decay.

Table 12
Training Details on ImageNet

Variants	T1, T2, T2d5, T3, T4, T5	T6	T8
Train Res	256		
Test Res	256		
Epochs	300	400	
Batch size	1024		
Optimizer	AdamW		
Momentum	0.9/0.999		
LR	3e-4		
LR decay	cosine		
Warmup epochs	20		
Warmup schedule	linear		
Label Smooth	0.1		
Mixup	0.8		
Cutmix	1.0		
Random Erasing	0.25		
Dropout	✗		0.01
EMA	✓		

A.2. Architectural Details

As WCANet incorporates wavelet transform mechanisms across multiple stages, the maximum downsampling factor of the network reaches 128. To ensure the complete alignment of feature dimensions and preserve the structural integrity of the wavelet decomposition, the input resolution must be perfectly divisible by this factor. Therefore, an input resolution of 256×256 is chosen as an intrinsic architectural requirement, avoiding the extensive feature padding, unnecessary computational overhead, and boundary artifacts that would inevitably arise from using a standard 224×224 resolution.

Tab.13 presents the architectural details of the three-stage WCANet variants, and Tab.14 shows the details of the four-stage WCANet variants.

A.3. The code of LWSA

To present the implementation of LWSA more intuitively, we provide the PyTorch code for LWSA. The specific code is shown in Listing 9. Due to space limitations, some functions are not fully displayed.

A.4. Limitation and Future Work

Limitation. While WCANet achieves a favorable speed-accuracy trade-off, our current evaluation focuses on GPU and CPU platforms, with dedicated mobile hardware deployment remaining a primary next step. Additionally, the current stage-dependent kernel size schedule ($k = 7, 5, 3$) is fixed, which may not optimally adapt to the varying spatial frequency distributions across highly diverse visual tasks. Finally, as a pure CNN, our model's performance in complex

Table 13
Architectural details of WCANet-T1, WCANet-T3, and WCANet-T5.

Stage	Resolution	Type	Config	WCANet		
				T1	T3	T5
stem	$\frac{H}{2} \times \frac{W}{2}$	Convolution	channels	8	13	16
	$\frac{H}{4} \times \frac{W}{4}$	Convolution	channels	16	26	32
	$\frac{H}{8} \times \frac{W}{8}$	Convolution	channels	32	52	64
	$\frac{H}{8} \times \frac{W}{8}$	Convolution	channels	64	104	128
1	$\frac{H}{16} \times \frac{W}{16}$	CCSA Block	channels	64	104	128
			blocks	2	2	3
2	$\frac{H}{32} \times \frac{W}{32}$	CCSA Block	channels	128	208	256
			blocks	3	3	4
3	$\frac{H}{64} \times \frac{W}{64}$	CCSA Block	channels	256	416	512
			blocks	2	2	3

Table 14
Architectural details of WCANet-T8.

Stage	Resolution	Type	Config	WCANet-T8
stem	$\frac{H}{2} \times \frac{W}{2}$	Convolution	channels	24
	$\frac{H}{4} \times \frac{W}{4}$	Convolution	channels	48
	$\frac{H}{8} \times \frac{W}{8}$	Convolution	channels	96
1	$\frac{H}{8} \times \frac{W}{8}$	CCSA Block	channels	96
			blocks	2
2	$\frac{H}{16} \times \frac{W}{16}$	CCSA Block	channels	192
			blocks	3
3	$\frac{H}{32} \times \frac{W}{32}$	CCSA Block	channels	384
			blocks	4
4	$\frac{H}{64} \times \frac{W}{64}$	CCSA Block	channels	640
			blocks	5

downstream dense prediction tasks still lags slightly behind the latest ViT and SSM-based frameworks.

Future Work. We will validate the model's efficiency on edge devices and publish the results on GitHub. To further bridge the performance gap in downstream tasks, we will explore architectural fusion strategies that integrate WCANet with other paradigms without increasing the computational burden. Furthermore, we intend to investigate dynamic, multi-perspective feature learning strategies, drawing inspiration from recent advancements such as HMP-Former [79], to enhance the model's architectural flexibility and generalization capability.

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